

Calculus & Trigonometry Reference Guide

Compiled Reference Document · Exponent Standard

1 Calculus Working Methodology & Rules

- **Step 1: Global Constants Extraction** — Always factor out any overall constant multiplier from the integrand and keep it to the left of the integral symbol before evaluating or substituting.
- **Strict Exponential Notation for Variables** — Avoid radical signs ($\sqrt{\quad}$) and denominators containing variables. Keep all variables strictly in exponential form:
 - Use $x^{1/2}$ instead of radical forms
 - Use $x^{-1/2}$ instead of fractions with roots in the denominator
 - Use x^{-1} , x^{-2} , $(kx)^{-n}$ instead of division by variable expressions
- **Fractions for Pure Constants Only** — Rational fraction bars are reserved strictly for constant values (e.g., $\frac{2}{5}$, $-\frac{6}{5}$, $\frac{-3}{1} \cdot \frac{2}{5}$).
- **Explicit Step Cancellation** — Explicitly show the simplification of exponents during substitutions (e.g., $x^{-1/2} \cdot x^{1/2} = x^0 = 1$).

2 Trigonometric Derivatives & Integrals

2.1 Basic Trigonometric Functions

Function $f(x)$	Derivative $\frac{d}{dx}[f(x)]$	Indefinite Integral $\int f(x) dx$
$\sin(x)$	$\cos(x)$	$-\cos(x) + C$
$\cos(x)$	$-\sin(x)$	$\sin(x) + C$
$\tan(x)$	$\sec^2(x)$	$\ln \sec(x) + C = -\ln \cos(x) + C$
$\csc(x)$	$-\csc(x) \cot(x)$	$-\ln \csc(x) + \cot(x) + C$
$\sec(x)$	$\sec(x) \tan(x)$	$\ln \sec(x) + \tan(x) + C$
$\cot(x)$	$-\csc^2(x)$	$\ln \sin(x) + C$

2.2 Common Trigonometric Integral Forms

Integrand	Result
$\int \sec^2(x) dx$	$\tan(x) + C$
$\int \csc^2(x) dx$	$-\cot(x) + C$
$\int \sec(x) \tan(x) dx$	$\sec(x) + C$
$\int \csc(x) \cot(x) dx$	$-\csc(x) + C$
$\int \sin^2(x) dx$	$\frac{1}{2}x - \frac{1}{4}\sin(2x) + C$
$\int \cos^2(x) dx$	$\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$

3 Common Trigonometric Values (Unit Circle)

3.1 Special Angles (First Quadrant)

Deg	Rad (θ)	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$	$\csc(\theta)$	$\sec(\theta)$	$\cot(\theta)$
0°	0	0	1	0	Undefined	1	Undefined
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{1}{2} \cdot 3^{1/2}$	$\frac{1}{3} \cdot 3^{1/2}$	2	$\frac{2}{3} \cdot 3^{1/2}$	$3^{1/2}$
45°	$\frac{\pi}{4}$	$\frac{1}{2} \cdot 2^{1/2}$	$\frac{1}{2} \cdot 2^{1/2}$	1	$2^{1/2}$	$2^{1/2}$	1
60°	$\frac{\pi}{3}$	$\frac{1}{2} \cdot 3^{1/2}$	$\frac{1}{2}$	$3^{1/2}$	$\frac{2}{3} \cdot 3^{1/2}$	2	$\frac{1}{3} \cdot 3^{1/2}$
90°	$\frac{\pi}{2}$	1	0	Undefined	1	Undefined	0

3.2 Quadrantal Angles & Quadrant Signs (ASTC Rule)

- $\theta = 0 \implies (x, y) = (1, 0), \quad \tan(0) = 0$
- $\theta = \frac{\pi}{2} \implies (x, y) = (0, 1), \quad \tan(\pi/2) \text{ is Undefined}$
- $\theta = \pi \implies (x, y) = (-1, 0), \quad \tan(\pi) = 0$
- $\theta = \frac{3\pi}{2} \implies (x, y) = (0, -1), \quad \tan(3\pi/2) \text{ is Undefined}$
- **Quadrant I** ($0 < \theta < \frac{\pi}{2}$): All positive (+)
- **Quadrant II** ($\frac{\pi}{2} < \theta < \pi$): Sine and Csc positive (+)
- **Quadrant III** ($\pi < \theta < \frac{3\pi}{2}$): Tangent and Cot positive (+)
- **Quadrant IV** ($\frac{3\pi}{2} < \theta < 2\pi$): Cosine and Sec positive (+)

4 Exponential & Logarithmic Functions

4.1 Derivatives

Function $f(x)$	Derivative $f'(x)$	Chain Rule Form $\frac{d}{dx}[f(u)]$
e^x	e^x	$u' \cdot e^u$
a^x ($a > 0, a \neq 1$)	$a^x \ln(a)$	$u' \cdot a^u \ln(a)$
$\ln(x)$	x^{-1}	$u' \cdot u^{-1}$
$\log_a(x)$	$(\ln a)^{-1} \cdot x^{-1}$	$u' \cdot (\ln a)^{-1} \cdot u^{-1}$

4.2 Integrals

Integrand	Indefinite Integral	Technique / Note
$\int e^x dx$	$e^x + C$	Standard
$\int e^{kx} dx$	$\frac{1}{k}e^{kx} + C$	u -substitution
$\int a^x dx$	$(\ln a)^{-1}a^x + C$	Exponential base a
$\int x^{-1} dx$	$\ln x + C$	Power -1 rule
$\int \ln(x) dx$	$x \ln(x) - x + C$	Parts: $u = \ln x, dv = dx$
$\int \log_a(x) dx$	$(\ln a)^{-1}(x \ln x - x) + C$	Change of base

4.3 Common Integration by Parts Combinations

- $\int x e^x dx = e^x(x - 1) + C$
- $\int x^2 e^x dx = e^x(x^2 - 2x + 2) + C$
- $\int x \ln(x) dx = \frac{1}{2}x^2 \ln(x) - \frac{1}{4}x^2 + C$
- $\int e^x \sin(x) dx = \frac{1}{2}e^x(\sin x - \cos x) + C$
- $\int e^x \cos(x) dx = \frac{1}{2}e^x(\cos x + \sin x) + C$

5 Inverse Trigonometric Functions (Arcs)

5.1 Derivatives of Inverse Trig Functions

Function $f(x)$	Derivative $f'(x)$	Chain Rule Form $\frac{d}{dx}[f(u)]$
$\arcsin(x)$	$(1 - x^2)^{-1/2}$	$u' \cdot (1 - u^2)^{-1/2}$
$\arccos(x)$	$-(1 - x^2)^{-1/2}$	$-u' \cdot (1 - u^2)^{-1/2}$
$\arctan(x)$	$(1 + x^2)^{-1}$	$u' \cdot (1 + u^2)^{-1}$
$\text{arccot}(x)$	$-(1 + x^2)^{-1}$	$-u' \cdot (1 + u^2)^{-1}$
$\text{arcsec}(x)$	$ x ^{-1}(x^2 - 1)^{-1/2}$	$u' \cdot u ^{-1}(u^2 - 1)^{-1/2}$
$\text{arccsc}(x)$	$- x ^{-1}(x^2 - 1)^{-1/2}$	$-u' \cdot u ^{-1}(u^2 - 1)^{-1/2}$

